

What is Software Engineering ?

A typical software project



How the customer explained it

A typical software project



How the project leader understood it

A typical software project



How the analyst designed it

A typical software project



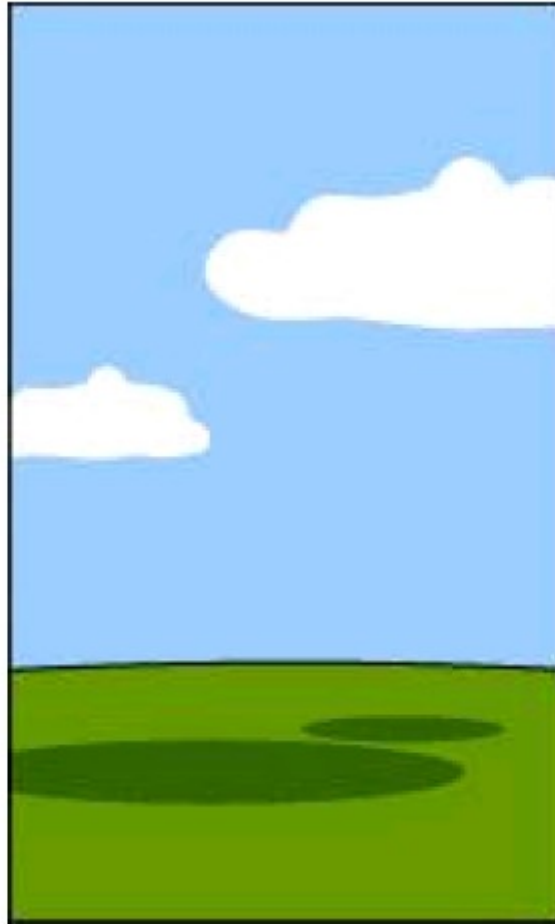
How the programmer wrote it

A typical software project



How the business consultant described it

A typical software project



How the project was documented

A typical software project



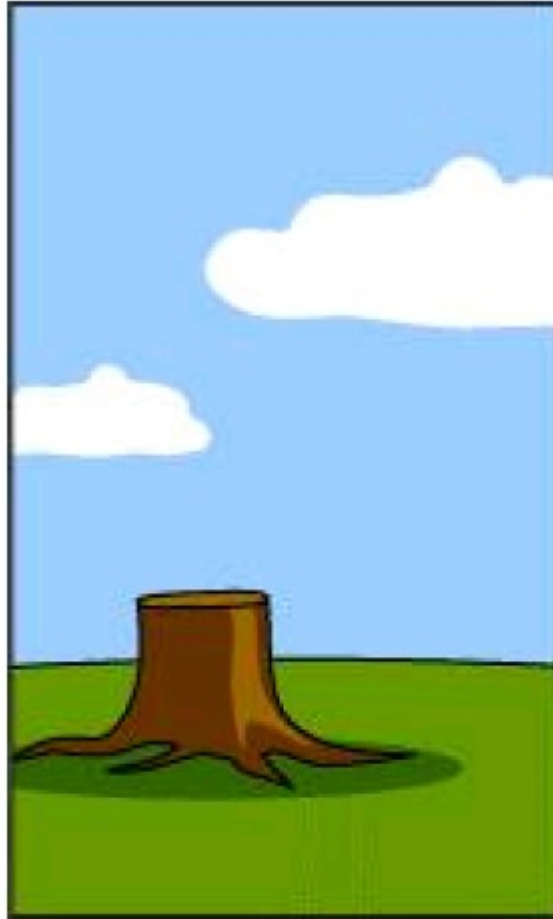
What operations installed

A typical software project



How the customer was billed

A typical software project



How it was supported

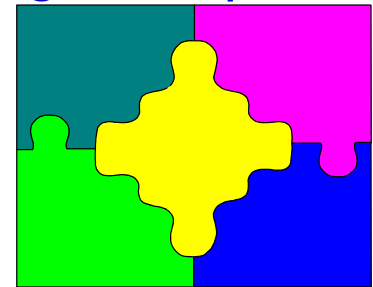
A typical software project



Finally, what the customer really needed

Why Study Software Engineering? (1)

- To acquire skills to develop large programs.
 - Exponential growth in complexity and difficulty level with size.
 - The ad hoc approach breaks down when size of software increases.
- Ability to solve complex programming problems.
 - How to break large projects into smaller and manageable parts?
 - How to use abstraction?
- Also learn techniques of,
 - Specification, design, user interface development, testing, project management, etc.
- **To acquire skills to be a better programmer.**
 - Higher Productivity and Better Quality Programs



Programs versus Software Products

• Usually small in size	• Large
• Author himself is sole user	• Large number of users
• Single developer	• Team of developers
• Lacks proper user interface	• Well-designed interface
• Lacks proper documentation	• Well documented & user-manual prepared
• Ad hoc development.	• Systematic development

Software Crisis

- Software products:
 - Fail to meet user requirements.
 - Frequently crash.
 - Expensive.
 - Difficult to alter, debug, and enhance.
 - Often delivered late.
 - Use resources non-optimally.

Factors Contributing to Software Crisis

- Larger problems,
- Lack of adequate training in software engineering,
- Increasing skill shortage,
- Low productivity improvements.